SLIDE 01 | PERSONALIZED GLUCOSE DIGITAL TWIN FOR TYPE 2 DIABETES

Leak-Free Ambulatory Glucose Forecasting Framework & Participant-Specific Calibration Engine

Hackathon Submission Details

- **Team:** DMounas Lab (Debangshu Mounas, Solo Developer & Researcher)
- **Affiliation:** National Institute of Technology Silchar (NIT Silchar)
- **Track:** Healthcare AI & Continuous Biomarker Intelligence
- **Repository:** DMounas/personalized-glucose-digital-twin
- **License:** MIT Open Source License

Executive Summary

- Evaluated across 45 participants from PhysioNet CGMacros cohort.
- Enforces strict 5-min history guard band against telemetry leakage.
- **+11.85% population skill** over persistence at both 30m & 60m horizons.
- **6-day personalization:** -1.81 mg/dL (8.0%) error reduction ($p < 0.05$, 40/45 improved).
- Interactive Streamlit dashboard in dual-tone clinical palette.

SLIDE 02 | TEAM & INSTITUTIONAL AFFILIATION

DMounas Lab — National Institute of Technology Silchar

Debangshu Mounas (Solo)

- **Role:** Lead Machine Learning Researcher & Full-Stack System Architect.
- **Affiliation:** National Institute of Technology Silchar (NIT Silchar).
- **GitHub Profile:** @DMounas (https://github.com/DMounas).
- **Focus:** Rigorous ML validation, leakage countermeasures, digital twins.

Core Values & Discipline

- **Research Integrity:** Preregistered protocols, zero data snooping.
- **Open Source:** Permissive MIT license with clear PhysioNet attribution.
- **Safety First:** Prominent non-clinical disclaimers and PHI privacy.
- **Quality Gates:** 135 automated tests, Ruff, and multi-version CI.

SLIDE 03 | THE PROBLEM: WHY AMBULATORY GLUCOSE ML FAILS

Three Structural Vulnerabilities in Published Glucose Forecasting Literature

Literature Pitfalls

- **1. Telemetry Interpolation Bleed:** Sensors log every 5 min; resampling to 1 min leaks future packets backwards.
- **2. Intra-Participant Contamination:** Random K-fold CV mixes readings from the same subject across train/test folds.
- **3. High Inter-Subject Heterogeneity:** Population models fail to capture individual insulin sensitivity and hepatic output.

Digital Twin Solutions

- **Strict 5-Minute Guard Band:** Blackout buffer [t-5, t] eliminating interpolation bleed.
- **Participant-Disjoint CV:** 50 folds with 0% intra-subject identity bleed.
- **Online Personalization:** Adapts global models using early patient telemetry.

SLIDE 04 | HEALTHCARE USE CASE & CLINICAL MOTIVATION

Empowering Proactive Diabetes Management & Clinical Decision Support

Clinical Value Chain

- **Proactive Lead Time:** 30 to 60 minutes advance warning before hyper/hypoglycemia occurs.
- **Actionable Window:** Enables timely patient action (light exercise, hydration, meal adjustments).
- **Clinician Support:** Quantifies individual trajectory confidence and residual variance.

Target Population

- **Type 2 Diabetes:** Mitigates postprandial excursions and glycemic variability.
- **Prediabetes:** Delivers dietary biofeedback to prevent progression.
- **Ambulatory Monitoring:** Tailored to free-living ambulatory individuals.

SLIDE 05 | WHY PHYSIOLOGICAL GLUCOSE FORECASTING MATTERS

Moving from Reactive Monitoring to Predictive Guidance

Global Diabetes Burden

- Over 537 million adults live with diabetes worldwide (IDF Diabetes Atlas).
- Glycemic variability drives severe microvascular and macrovascular complications.
- Commercial CGMs are reactive: alerts trigger only after thresholds are crossed.

Digital Twin Impact

- Simulates individual patient glycemic dynamics based on recent telemetry and meals.
- Bridges the gap between static guidelines and personalized daily reality.
- Enables what-if timing evaluations to optimize metabolic control.

SLIDE 06 | DATASET: PHYSIONET CGMACROS COHORT

Open-Access Clinical Telemetry (Bent et al., Scientific Data 2025)

Cohort Composition

- **45 diverse subjects** across the metabolic health continuum.
- **Subgroups:** Normal Controls (15), Prediabetes (16), Type 2 Diabetes (14).
- Telemetry logged via Dexcom G6 continuous monitors.
- Demographics: Age, Sex, BMI, and baseline HbA1c.

Data Quality & Integrity

- **Delta t = 60s rule:** Contiguous run segmentation with zero synthetic interpolation.
- **Lattice Grid:** m(t) = round((t-t0)/60s), issuance on m(t) mod 5 == 0.
- **123,599 valid origins** evaluated across the full cohort.

SLIDE 07 | END-TO-END SYSTEM ARCHITECTURE

Modular Five-Layer Pipeline from Raw Sensor Ingestion to User Interface

Feature Engineering Space

- **Ingestion:** Run segmentation, lattice grid alignment (phi = 0).
- **Guarded Features:** 13 autoregressive lags [y(t-65) ... y(t-5)].
- **Multimodal Context:** 3 meal-timing flags + 4 clinical covariates (R^20 space).

Modeling & Delivery

- **Global Ladder:** 8 rungs evaluated under 50 disjoint folds.
- **Personalization:** Regularized Ridge residual calibration on early study days.
- **Interactive UI:** Modular Streamlit dashboard with zero raw data dependencies.

SLIDE 08 | FIVE ARCHITECTURAL LEAKAGE COUNTERMEASURES

Eliminating Temporal Bleed & Intra-Subject Memorization

Temporal Protections

- **1. 5-Min Guard Band Gap:** Mandatory zero-input buffer in [t-5, t].
- **2. 13 Permitted Lags:** Strictly bounded to [y(t-65) ... y(t-5)].
- **3. 120-Min Purge Buffer:** Mandatory blackout separating adaptation from test days.

Evaluation Protections

- **4. Participant-Disjoint CV:** 10 repeats x 5 folds with zero subject overlap.
- **5. Participant Bootstrap:** B=2,000 resamples of 45 independent participant units.
- **Guarantee:** Zero test subject telemetry or covariates seen during training.

SLIDE 09 | THE STAGE 2 GLOBAL FORECASTING LADDER

Progressive 8-Rung Benchmark on Identical Origins (N=123,599)

Rungs 0 through 4b

- **Rung 0:** Guarded Persistence baseline (latest lag y(t-5)).
- **Rung 1:** Damped Trend Extrapolation (velocity damped by gamma).
- **Rung 2:** Guarded Ridge Regression (13 autoregressive lags).
- **Rung 3 & 4:** Activity & Meal timing context.

Advanced Rungs 5 & 6

- **Rung 5:** Clinical Covariates Ridge (Age, Sex, BMI, HbA1c in R^20 space).
- **Rung 6:** Fixed HistGradientBoostingRegressor on identical R^20 features.
- Evaluated across strictly identical evaluation origins.

SLIDE 10 | STAGE 2 GLOBAL FORECASTING RESULTS

Rung 5 Ridge Achieves Best Population Performance & Parity with GBDT

Headline Benchmark Metrics

- **Persistence (R0):** 21.55 mg/dL (+30m) | 29.32 mg/dL (+60m).
- **Autoregressive Core (R2):** 19.87 mg/dL (+30m) | 26.97 mg/dL (+60m).
- **Selected Foundation (R5): 18.85 mg/dL** (+30m) | **25.45 mg/dL** (+60m).
- **Population Skill: +11.82%** (+30m) | **+11.85%** (+60m) vs Persistence.

Ridge vs. GBDT Parity

- GBDT (Rung 6): 19.15 mg/dL (+30m) | 25.64 mg/dL (+60m).
- Paired Delta-RMSE between GBDT and Ridge crosses zero at both horizons.
- GBDT did not demonstrate a statistically significant gain over Ridge.
- Ridge provides closed-form stability and instant training.

SLIDE 11 | STAGE 3 PERSONALIZATION METHODOLOGY

Individual Adaptation via Regularized Residual Correction

Experimental Design

- **Adaptation Windows:** Evaluates first 2, 4, and 6 study days.
- **Purge Blackout:** Mandatory >= 120-min buffer separating adaptation from test.
- **Matched Baseline:** Evaluated on strictly identical post-purge origins.

Residual Calibration

- **Foundation:** Pre-trained Rung 5 Ridge global model.
- **Residual Target:** r(t) = y(t+h) - y_hat_global(t+h).
- **Model:** Ridge fitted on 16 dynamic features (13 lags + 3 meal timing).
- **Evaluation:** 100% on unseen subsequent telemetry.

SLIDE 12 | STAGE 3 PERSONALIZATION RESULTS

Statistically Significant Error Reduction Under 6-Day Adaptation ($p < 0.05$)

+60-Minute Horizon Impact

- **2 Study Days:** Delta-RMSE = -0.025 mg/dL (spans zero; sparse telemetry).
- **4 Study Days:** Delta-RMSE = -1.146 mg/dL ($p < 0.05$, strictly excludes zero).
- **6 Study Days: -1.811 mg/dL** [-2.43, -1.09] (**8.0% relative reduction**).
- **Cohort Benefit: 40 out of 45 participants (88.9%) improve.**

Subgroup Excursion Dynamics

- **Normal Controls:** -1.33 mg/dL error change.
- **Prediabetes:** -1.82 mg/dL error change.
- **Type 2 Diabetes: -2.34 mg/dL error change** (largest absolute therapeutic benefit).

SLIDE 13 | INTERACTIVE RESEARCH DASHBOARD TOUR

Light Dual-Tone Clinical Interface Built with Streamlit & Plotly

Four Dedicated Modules

- **1. Project Overview:** Cohort summary, ML failure modes, system architecture.
- **2. Global Explorer:** Horizon toggle (+30m, +60m), 8-rung benchmark ladder with 95% CIs.
- **3. Personalization Lab:** Window controls (2d, 4d, 6d), dose-response plots, subgroup table.
- **4. Methodology:** Guard-band schematic, protocol guide, responsible use.

Design Discipline

- Mild green and white dual-tone clinical palette with **zero emojis**.
- Powered by verified canonical aggregate metrics — zero raw data exposure.
- Instant local launch: streamlit run app.py.

SLIDE 14 | TECHNICAL ARCHITECTURE & REPOSITORY HYGIENE

Production-Grade Research Engineering Standards

Core Technologies

- **Languages:** Python 3.10, 3.11, 3.12 (native multi-version verification).
- **ML Core:** Scikit-learn (Ridge, HistGradientBoosting, GroupKFold).
- **Data & Math:** NumPy, Pandas, PyYAML.
- **UI & Visualization:** Streamlit, Plotly, Matplotlib.

Quality Gates & CI/CD

- **135 automated unit & regression tests** passing cleanly (pytest).
- Strict linting & formatting enforced by Ruff (0 violations).
- Complete static type safety verified via Mypy (0 issues).
- GitHub Actions CI matrix testing every commit.

SLIDE 15 | LIMITATIONS & RESPONSIBLE AI FRAMEWORK

Scientific Rigor, Observational Boundaries & Clinical Disclaimers

Methodological Boundaries

- **Observational Data:** Observational telemetry does not establish causal physiological mechanisms.
- **No Closed-Loop Control:** Benchmarks forecasting accuracy; does not titrate insulin.
- **Hardware Drift:** Does not simulate acute sensor displacement or compression artifacts.

Responsible AI Standards

- **Not FDA-cleared:** Research prototype strictly intended for benchmarking and education.
- Not a diagnostic or treatment system; never substitute for clinical advice.
- **Zero PHI:** Complete privacy compliance without raw patient data exposure.

SLIDE 16 | CONCLUSION & STRATEGIC IMPACT

Key Takeaways from the Glucose Digital Twin Benchmark

Summary of Contributions

- **1. Defensible Methodology:** Established the 5-min guard band and temporal purge standard.
- **2. Benchmark Foundation:** Proven that Rung 5 Ridge achieves +11.8% skill with mathematical stability.
- **3. Personalization Proof:** 6-day adaptation delivers a statistically significant 8.0% error drop ($p < 0.05$).
- **4. Clinical Equity:** Largest benefit (-2.34 mg/dL) achieved in Type 2 Diabetes cohort.

Future Horizons

- Wearable integration (continuous physical activity and meal photologging).
- Nocturnal hypoglycemia risk prediction across multi-day circadian windows.
- Federated on-device model adaptation preserving patient privacy at the edge.

SLIDE 17 | REPOSITORY, DEMO & SUBMISSION LINKS

All Artifacts Are Publicly Accessible and Fully Reproducible

Public Submission Resources

- **GitHub Repository:** https://github.com/DMounas/personalized-glucose-digital-twin
- **Architecture Diagram:** docs/submission/architecture.pdf | architecture.pptx
- **Presentation Slide Deck:** docs/submission/presentation.pdf | presentation.pptx
- **Demo Video Script:** docs/submission/DEMO_SCRIPT.md

Open Source & Attribution

- **License:** MIT License (Copyright 2026 Debangshu Mounas).
- **Dataset Attribution:** PhysioNet CGMacros 1.0.0 (Bent et al., Scientific Data 2025).
- **Local Launch:** streamlit run app.py
- Thank you for reviewing the DMounas Lab Hackathon Submission!